

DUARTE FONSECA

SOFTWARE ENGINEER & GAME DEVELOPER

CONTACTS:

 duarte.mfonseca@gmail.com



[Portfolio](#)



[Github](#)



[Linkedin](#)



[itch.io](#)

SUMMARY

Junior game developer and software engineer with a focus in gameplay systems, procedural generation, and performance optimization, with experience across design, programming, and 3D modeling.

WORK EXPERIENCE

Teacher at Mad Fox Labs

August, 2025 – Present

- Taught extracurricular STEM modules focused on Godot Game Development, Robotics, and Artificial Intelligence.

EDUCATION

Graduation: July 2026

Bachelor of Software Engineering and Multimedia - Lisbon School of Engineering (ISEL)

- Specialization in Game Development (24 ECTS)

PROJECTS

Minecraft Clone - Unity

January 2026 - February 2026

- Built infinite, optimized voxel world generation using FrBM and chunk-based rendering
- Implemented efficient mesh generation with face culling and custom UV mapping

Zombie Meltdown - Unity

July 2025 - December 2025

[Link to learn More about this project](#)

[Gameplay Showcase](#)

- Designed modular weapon and animation systems using Scriptable Objects
- Optimized enemy updates using a centralized update loop, reducing MonoBehaviour overhead and enabling large-scale enemy waves.

Space Game - Java + Processing (No Engine)

December 2024 - January 2025

[Link to learn more about this project](#)

- Engineered a custom physics engine with collision detection, rigid-body dynamics, and particle systems
- Implemented procedural cave generation using cellular automata and custom particle systems

EXTRACURRICULUM

ISEL Game Studios

March 2025 - Present

- Manage staff and oversee event production, fostering an active student game dev community
- Founded and lead a 95-member game development club, organizing game jams, talks, and collaborative projects

GameDev Técnico

January 2024 - Present

- Organize gaming events and social gatherings to strengthen community interaction
- Lead community management for a 244-member student club, ensuring engagement and a safe, inclusive environment

SKILLS

- Programming Languages: Python, C#, GDscript, Kotlin, Java, HTML, JavaScript, CSS, SQL
- Tools: Unity, Godot, Blender, Android Studio, Krita